

Partition Readers of Permutations

Exact Information Profiles, Hidden Fibers, and the Cost of Unbalanced Transitions

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Abstract

A permutation T of N cells is observed by a *partition reader*: the cells are cut into m blocks, and the reader records only how many cells each block sends into each block. We determine exactly how much such a reader learns, what remains hidden, and when two basic structural features, the direction of the transformation and its cycle structure, become visible.

The set of permutations consistent with an observed table is a double coset of a Young subgroup. The information seen plus the logarithm of the size of this fiber equals $\log_2 N!$ for every single observation, and the same identity holds in expectation. For m equal blocks of size s we prove the exact average information

$$I(T; C_m) = \log_2 N! - 2m \log_2 s! + m^2 E[\log_2 c!],$$

where c is hypergeometric with parameters (N, s, s) . The formula agrees with exhaustive enumeration to six decimal places.

A reader with one cut sees only the number of cells crossing it. It learns $1/2 \log_2(\pi e N/8) + o(1)$ bits, about 3.55 bits for $N = 128$, out of $\log_2 128! \approx 716$, and it can never distinguish T from T^{-1} . Direction becomes readable at three blocks. For cycles we prove a sharper statement: whenever some block contains two or more cells, no table determines the number of cycles. The cycle counts within each fiber form an interval of consecutive integers whose minimum is the number of connected components of the block graph, so only the full-resolution reader sees the cycle structure.

Finally, we treat maps that are not bijections. The block table alone fixes the exact minimum amount of information any compatible map must erase. That minimum is zero precisely when the table is balanced, and every forced excess costs at least two bits per N cells. By Landauer's principle this sets an energy floor determined by the table.

All statements are verified exhaustively in the stated ranges. Two transformations driven by the hexadecimal digits of π and by a SHA-256 chain illustrate the profile at $N = 128$. A self-contained notebook reproduces every number.

Keywords: permutations, set partitions, contingency tables, double cosets, hypergeometric distribution, mutual information, cycle structure, Landauer’s principle.

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1. Introduction

Let T be a permutation of a set of N cells. Two numbers set the scale of the problem. Naming a single cell takes $\log_2 N$ bits, and naming the whole transformation takes $\log_2 N!$ bits. For $N = 128$ these are 7 and approximately 716.16.

An observer who cannot follow individual cells, but can see which region each cell came from and which region it went to, lies somewhere between these two scales. This paper determines exactly where.

We model such an observer as a *partition reader*. The cells are divided into blocks, and the reader returns the table whose (i, j) entry counts the cells of block i that T sends into block j . Two permutations give the same table if and only if they lie in the same double coset of the Young subgroup of the partition $[2, 4]$. The reader therefore sees the double coset and nothing more.

Main results.

1. **Accounting.** For every observed table C , the bits seen, $\log_2 N! - \log_2 |\mathcal{F}(C)|$, plus the hidden bits, $\log_2 |\mathcal{F}(C)|$, equal $\log_2 N!$. Averaged over uniform T , the seen bits equal the mutual information $I(T; C_m)$. Finer readers have nested fibers and never see less (Section 3).
2. **Exact information profile.** For equal blocks, $I(T; C_m)$ has the closed form stated in the abstract (Theorem 4.1). For one cut it reduces to the entropy of a hypergeometric variable, asymptotically $1/2 \log_2(\pi e N/8)$ (Corollary 4.2).
3. **Direction.** Inversion transposes the table. A reader with two blocks is always blind to direction; three blocks suffice to see it; at full resolution the reader is blind exactly on involutions (Section 5).
4. **Cycles.** If some block has at least two cells, no table fixes the number of cycles; the cycle counts within a fiber form an interval whose minimum is the number of components of the block graph (Theorem 6.1).
5. **Unbalanced transitions.** For arbitrary maps, the table determines the exact minimum information erased; that minimum vanishes exactly for balanced tables, and each forced excess costs at least $2/N$ bits per cell (Section 7).

Related work. The correspondence between double cosets of Young subgroups and contingency tables, and the Fisher–Yates formula for double-coset sizes, are classical [2, 4]. The distribution of cycle type within such double cosets is treated in [5]. Theorem 6.1(c) uses its characterization of the minimum number of cycles. Mutual information and the chain rule are standard [1]. The thermodynamic reading of Section 7 rests on Landauer’s principle [6], on Bennett’s analysis of reversible computation [7], and on the experimental

verification of the Landauer bound [8]. Uniform shuffling by the Fisher–Yates algorithm is described in [9], and SHA-256 is specified in [10].

2. Setup

Throughout, $N \geq 1$ and the cells are $0, 1, \dots, N - 1$. For a divisor m of N put $s = N/m$.

Definition 2.1 (reader). The reader at depth m cuts the cells into the m contiguous blocks $B_i = \{is, \dots, (i+1)s - 1\}$. For a permutation T it returns the $m \times m$ table $C_m(T)$ whose entries are

$$C_m(T)_{ij} = |\{x \in B_i: T(x) \in B_j\}|.$$

Every row and every column of $C_m(T)$ sums to s . When $m = 2^k$ we say the reader has k cuts.

Definition 2.2 (fiber). For a table C , the fiber is $\mathcal{F}(C) = \{T' \in S_N: C_m(T') = C\}$.

Definition 2.3 (seen and hidden bits). For a permutation T and depth m ,

$$\text{hidden}_m(T) = \log_2 |\mathcal{F}(C_m(T))|, \quad \text{seen}_m(T) = \log_2 N! - \text{hidden}_m(T).$$

When T is uniformly distributed on S_N , $\text{seen}_m(T)$ is the surprisal of the observed table, $-\log_2 \Pr[C_m = C_m(T)]$.

3. Fibers

Lemma 3.1 (fiber size). For a table C with row and column sums s ,

$$|\mathcal{F}(C)| = \frac{(s!)^{2m}}{\prod_{i,j} C_{ij}!}.$$

Proof. The fiber is the double coset KTK of the Young subgroup $K = \prod_i S_{B_i}$, of order $(s!)^m$. Its size is $|K|^2/|K \cap TKT^{-1}|$. The intersection consists of the permutations that permute each set $B_i \cap T(B_j)$ among itself, and $B_i \cap T(B_j)$ has C_{ji} elements. Hence $|K \cap TKT^{-1}| = \prod_{i,j} C_{ij}!$ [4]. ◻

Proposition 3.2 (accounting). For every T , $\text{seen}_m(T) + \text{hidden}_m(T) = \log_2 N!$. If T is uniform, then

$$I(T; C_m) = H(C_m) = \mathbb{E}[\text{seen}_m(T)], \quad I(T; C_m) + \mathbb{E} \log_2 |\mathcal{F}(C_m)| = \log_2 N!.$$

Proof. The first identity is the definition. Since C_m is a function of T , $I(T; C_m) = H(C_m)$. Given $C_m = C$, the permutation T is uniform on $\mathcal{F}(C)$, so $H(T | C_m) = \mathbb{E} \log_2 |\mathcal{F}(C_m)|$. The chain rule $H(T) = I(T; C_m) + H(T | C_m)$ [1], together with $H(T) = \log_2 N!$, gives the rest. Finally, $\Pr[C_m = C] = |\mathcal{F}(C)|/N!$, so the expected surprisal is $H(C_m)$. ◻

Proposition 3.3 (nesting). Let $m \mid m'$, so that every block of the depth- m' reader lies inside a block of the depth- m reader. Then $C_{m'}(T)$ determines $C_m(T)$ by summing entries, $\mathcal{F}(C_{m'}(T)) \subseteq \mathcal{F}(C_m(T))$, $\text{seen}_{m'}(T) \geq \text{seen}_m(T)$ for every T , and $I(T; C_{m'}) \geq I(T; C_m)$. At $m = N$ the fiber is $\{T\}$ and the reader sees all $\log_2 N!$ bits.

Proof. Summing the depth- m' table over groups of blocks gives the depth- m table. The remaining statements follow from the inclusion of fibers. ▫

Example 3.4. Take $N = 128$ and $m = 64$, so every block contains two cells. By Lemma 3.1,

$$\text{hidden}_{64}(T) = 128 - a(T),$$

where $a(T)$ is the number of entries of the table equal to 2. That is, it counts the blocks that send both of their cells into a single block. For the π -driven transformation of Section 8, $a = 1$ and 127 bits are hidden. For the SHA-256-driven transformation, $a = 0$ and 128 bits are hidden.

4. The information profile

Theorem 4.1. Let T be uniform on S_N and let $m \mid N$ with $s = N/m$. Then

$$I(T; C_m) = \log_2 N! - 2m \log_2 s! + m^2 E[\log_2 c!], \quad c \sim \text{Hyp}(N, s, s),$$

where $\text{Hyp}(N, s, s)$ is the hypergeometric distribution: the number of marked items among s drawn without replacement from N items of which s are marked.

Proof. By Proposition 3.2 and Lemma 3.1,

$$I(T; C_m) = \log_2 N! - 2m \log_2 s! + \sum_{i,j} E[\log_2 C_{ij}!].$$

For fixed i and j , the entry C_{ij} counts how many of the s cells of B_i a uniform permutation sends into the s positions of B_j . The image of B_i is a uniformly random s -subset of the N positions, so $C_{ij} \sim \text{Hyp}(N, s, s)$ for every pair. Linearity of expectation over the m^2 entries completes the proof. ▫

The formula agrees with exhaustive enumeration of all permutations for $N = 8$ (at $m = 1, 2, 4, 8$) and $N = 9$ (at $m = 1, 3, 9$) to six decimal places. Table 1 compares it with sampling at $N = 128$.

Table 1. The information profile at $N = 128$, in bits. The total is $\log_{2128}! = 716.162$.

cuts	m	Theorem 4.1	mean of 2,000 uniform permutations
0	1	0.000	0.000
1	2	3.553	3.551
2	4	25.936	25.980
3	8	101.325	101.331
4	16	250.332	250.336
5	32	427.240	427.229
6	64	588.666	588.675
7	128	716.162	716.162

Corollary 4.2 (the first reader). At $m = 2$ the table has the form $\binom{s-c}{c} \binom{c}{s-c}$ with $s = N/2$. It is determined by the single crossing number $c \sim \text{Hyp}(N, N/2, N/2)$, and

$$I(T; C_2) = H(c) = 1/2 \log_2 \left(\frac{\pi e N}{8} \right) + o(1) \quad (N \rightarrow \infty).$$

Proof. Equal row and column sums force the displayed form. Hence $H(C_2) = H(c)$. The variable c has variance $N/16 \cdot N/N - 1$. By the local central limit theorem for the hypergeometric distribution [3], its entropy is that of a discretized normal law with this variance up to $o(1)$, namely $1/2 \log_2(2\pi e N/16) + o(1)$. ▫

Table 2. Information seen by the one-cut reader.

N	exact $H(c)$	$1/2 \log_2(\pi e N/8)$
8	1.6419	1.5471
64	3.0584	3.0471
128	3.5527	3.5471
1,024	5.0478	5.0471
65,536	8.0471	8.0471
1,048,576	10.0471	10.0471

The profile rises steeply between the first and last cuts (Figure 1). Half of $\log_2 N!$ is first reached at cut 3, 4, 4, 5, 6, 6 and 7 for $N = 16, 32, \dots, 1024$ (Appendix A). Figure 2 shows the profiles normalized by $\log_2 N$ cuts and $\log_2 N!$ bits.

Figure 1. Bits seen at each depth for $N = 128$: the exact average of Theorem 4.1 and the two hex-driven transformations of Section 8.

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Figure 2. Normalized information profiles for $N = 16, 64, 256$ and $1,024$.

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5. Direction

Proposition 5.1. $C_m(T^{-1}) = C_m(T)^t$. Consequently, the reader at depth m distinguishes T from T^{-1} if and only if $C_m(T)$ is not symmetric.

Proof. $|\{x \in B_i : T^{-1}(x) \in B_j\}| = |\{y \in B_j : T(y) \in B_i\}| = C_m(T)_{ji}$. ▫

Theorem 5.2.

- (a) Every table at depth $m = 2$ is symmetric, so the one-cut reader never distinguishes a transformation from its inverse.
- (b) For every $m \geq 3$ with $m \mid N$ there is a permutation whose table is not symmetric.
- (c) At $m = N$ the reader fails to distinguish T from T^{-1} exactly when T is an involution.

Proof. (a) follows from the form displayed in Corollary 4.2. For (b), let T send block B_i onto block B_{i+1} for each i , with indices taken modulo m . Then $C_{i,i+1} = s$ and $C_{i+1,i} = 0$ when $m \geq 3$. For (c), the table at $m = N$ is the permutation matrix of T , which is symmetric exactly when $T = T^{-1}$. ▫

Table 3. Exact counts of permutations whose table is symmetric. At $m = N$ they are the involutions (10, 76, 764 and 2,620). In a sample of 20,000 uniform permutations of $N = 128$ cells, the fraction is 1.0000 at $m = 2$, 0.0037 at $m = 4$, and 0 at every $m \geq 8$.

N	m	permutations blind to direction	fraction
4	2	24 of 24	1.0000
4	4	10 of 24	0.4167
6	2	720 of 720	1.0000
6	3	384 of 720	0.5333
6	6	76 of 720	0.1056
8	2	40,320 of 40,320	1.0000
8	4	7,584 of 40,320	0.1881
8	8	764 of 40,320	0.0189
9	3	164,160 of 362,880	0.4524
9	9	2,620 of 362,880	0.0072

6. Cycles

Theorem 6.1. Let P be any partition of the cells into blocks of which at least one has two or more cells. Let C be a table of P and $\mathcal{F}(C)$ its fiber.

- (a) The number of cycles is not constant on $\mathcal{F}(C)$.
- (b) The set $\{\text{cyc}(T) : T \in \mathcal{F}(C)\}$ is an interval of consecutive integers.
- (c) Its minimum equals the number of connected components of the directed multigraph on the blocks with C_{ij} arcs from i to j .

Proof. Let K be the Young subgroup of P , so that $\mathcal{F}(C) = KTK$ for any T in the fiber.

- (a) Choose two cells $x \neq y$ in a common block and put $T' = T \circ (x y)$. The table does not change, because x and y lie in the same source block. Multiplying a permutation by a transposition changes its number of cycles by exactly one. So T and T' lie in the same fiber and have different cycle counts.
- (b) K is generated by transpositions of cells within a block. Every element of KTK is therefore reached from T by a sequence of left and right multiplications by such transpositions, and each step changes the cycle count by exactly one. A set of integers reachable in steps of size one is an interval.
- (c) This is Proposition 5.4 of [5]. ▫

Corollary 6.2. Among all partition readers, only the full-resolution reader, with every block a single cell, determines the number of cycles of the transformation. The same holds for its cycle type and, by the transposition argument, for its sign.

For $N = 8$ and $N = 9$ we checked all permutations and all depths. No table below full resolution fixes the number of cycles, the cycle counts in every fiber form an interval, and their minimum equals the number of components in every case. The mean certified minimum at $N = 8$ is 1.0143 at $m = 2$ and 1.1722 at $m = 4$, against a mean of 2.7179 actual cycles. The π -driven and SHA-256-driven transformations of Section 8 have 3 and 9 cycles; their tables certify one cycle at every depth below $m = 128$.

7. Unbalanced transitions

Now let $f: [n] \rightarrow [n]$ be an arbitrary map, not necessarily a bijection. Let P be a partition with block sizes $d_1, \dots, d_m$. The table of f has row sums d_j and column sums c_j , the numbers of arrivals into each block. With the input X uniform, the information erased by f is

$$E(f) = H(X | f(X)) = \frac{1}{n} \sum_y |f^{-1}(y)| \log_2 |f^{-1}(y)|.$$

For integers $c, d \geq 1$ define $\varphi(c, d) = 0$ if $c \leq d$. Otherwise write $c = qd + r$ with $0 \leq r < d$ and put

$$\varphi(c, d) = (d - r) q \log_2 q + r (q + 1) \log_2 (q + 1).$$

Put $D = \sum_j \max(0, c_j - d_j)$.

Theorem 7.1 (minimum erasure). The minimum of $E(f)$ over all maps f with a given table equals $\frac{1}{n} \sum_j \varphi(c_j, d_j)$.

Proof. $E(f)$ depends only on the fiber sizes of f . The table fixes how many arrivals each block receives but not which of its cells they reach, so the blocks can be filled independently. If $c_j \leq d_j$, the arrivals can reach distinct cells and contribute nothing. Otherwise the c_j arrivals are distributed over d_j cells. Since $x \log_2 x$ is convex, $\sum x \log_2 x$ over the fiber sizes is minimized by the most even distribution: $d_j - r$ fibers of size q and r of size $q + 1$. That minimum is $\varphi(c_j, d_j)$. ▀

Corollary 7.2. The minimum erasure is zero if and only if $c_j = d_j$ for every block, which is the balanced case. Balanced tables are exactly the tables of permutations, the maps that erase nothing.

Proof. Zero requires $c_j \leq d_j$ for all j . Since $\sum_j c_j = \sum_j d_j = n$, this forces equality. ▀

Corollary 7.3. Every map with a given table erases at least $2D/n$ bits, with equality for its minimizers whenever $c_j \leq 2d_j$ for all j .

Proof. On $d \leq c \leq 2d$ the function $\varphi(\cdot, d)$ is linear with slope 2, and it is convex for $c \geq d$. Hence $\varphi(c, d) \geq 2(c - d)$. ▀

Proposition 7.4. At most $n - D$ points of any map with the given table lie on cycles of f .

Proof. Points on cycles lie in the image of f . The image meets block j in at most $\min(c_j, d_j)$ points, and $\sum_j \min(c_j, d_j) = n - D$. ▀

Remark 7.5. By Landauer’s principle [6], erasing one bit of information dissipates at least $k_B T \ln 2$ of energy, which is 2.871×10^{-21} J at 300 K. The bound has been verified experimentally [8]. Logically reversible operations, the bijections, have no such minimum [7]. Theorem 7.1 therefore gives an energy floor for any physical process realizing a given block table, and that floor is computable from the table alone.

The statements of this section were checked over every map $f: [n] \rightarrow [n]$ and every block type for $n \leq 6$: 513,216 maps and 59,486 distinct tables at $n = 6$, with no exceptions.

8. Computation

Hex-driven transformations. To illustrate the profile on specific transformations, we generate permutations of $N = 128$ cells from two hexadecimal streams:

- the hexadecimal digits of π , computed exactly by Machin’s formula (the first 32 digits are checked against 243f6a8885a308d313198a2e03707344);
- the SHA-256 chain $h_0 = \text{SHA256}(\text{empty input})$, $h_{k+1} = \text{SHA256}(h_k)$ [10].

A stream is converted into a permutation by the Fisher–Yates algorithm [9]. Each draw uses one byte, masked to the needed number of bits, and draws outside the valid range are rejected, so the shuffle is unbiased when the bytes are.

Table 4. Seen and hidden bits for the two hex-driven transformations at $N = 128$. Both transformations become distinguishable from their inverses at two cuts.

cuts	π : seen	π : hidden	SHA-256: seen	SHA-256: hidden	direction visible
0	0.00	716.16	0.00	716.16	no
1	3.19	712.97	4.26	711.90	no
2	25.96	690.21	22.36	693.80	yes
3	104.05	612.11	105.49	610.68	yes
4	250.34	465.82	252.34	463.82	yes
5	427.72	288.44	426.72	289.44	yes
6	589.16	127.00	588.16	128.00	yes
7	716.16	0.00	716.16	0.00	yes

Verification. All computations use exact integer arithmetic or double-precision logarithms of factorials. Table 5 lists the checks. A self-contained Jupyter notebook, using only the Python standard library, reproduces every number in this paper and runs in about one minute.

Table 5. Computational verification.

Statement	Check	Range
Lemma 3.1, Proposition 3.3	fiber sizes by grouping all permutations; refinement determines coarser table	$N = 8, 9$, all depths
Theorem 4.1	formula against exact entropy of the table	$N = 8, 9$, all depths

Statement	Check	Range
Theorem 4.1	formula against sample mean	$N = 128, 2,000$ permutations
Corollary 4.2	exact entropy against asymptotic form	N up to 2^{20}
Proposition 5.1, Theorem 5.2	transpose law; symmetric-table counts; involutions	$N \leq 9$, all depths; $N = 128$ sampled
Theorem 6.1	cycle counts per fiber; interval property; minimum equals components	$N = 8, 9$, all depths
Theorem 7.1, Corollaries 7.2–7.3, Proposition 7.4	all maps and block types	$n \leq 6$

9. Discussion

A partition reader is the simplest observer that sees relations between regions without following individual cells. The results above describe what such an observer can know, and when.

The amount it learns is governed by one hypergeometric expectation (Theorem 4.1). The one-cut reader learns only the size of the exchange across its cut, which grows like $1/2 \log_2 N$. It learns nothing about direction, because two balanced blocks can only exchange symmetrically. Direction needs a third block. Cycle structure, the closed orbits of the transformation, is hidden at every resolution short of the full one (Theorem 6.1).

For maps that are not bijections, the same table measures what the map must forget: nothing when the table is balanced, and a computable positive amount otherwise.

The profiles here are computed under the uniform distribution on permutations. For structured transformations, the seen and hidden bits of a single observation are still given exactly by Lemma 3.1. The two hex-driven examples lie close to the uniform average at every depth.

10. Open problems

1. **Shape of the profile.** Determine the limiting shape of $I(T; C_{2^k})/\log_2 N!$ as a function of $k/\log_2 N$, and the depth at which half of $\log_2 N!$ becomes visible.
2. **Direction at intermediate depths.** Find the asymptotic probability that the table at depth $m \geq 3$ is symmetric. At $N = 128$ it is already 0.0037 at $m = 4$.
3. **Cycle spread.** Describe the distribution of the width of the cycle-count interval of Theorem 6.1 over fibers, as a function of the reader's depth.
4. **Iterated maps.** For a map iterated until every point lies in its recurrent set, determine the minimum total erasure in terms of the block table.
5. **Structured sources.** Characterize the transformations, such as those generated by specific hash functions or digit sequences, whose profiles deviate measurably from the uniform average.

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Appendix A. Exact profiles for $N = 16, \dots, 1024$

Values of $I(T; C_{2^k})$ in bits from Theorem 4.1, for cuts $k = 0, 1, \dots, \log_2 N$. The last value in each row is $\log_2 N!$.

Table A1. Exact information profiles.

N	profile by cuts	half reached at cut
16	0.00, 2.09, 12.18, 28.78, 44.25	3
32	0.00, 2.57, 16.86, 48.86, 86.18, 117.66	4
64	0.00, 3.06, 21.43, 74.21, 153.81, 232.50, 296.00	4
128	0.00, 3.55, 25.94, 101.33, 250.33, 427.24, 588.67, 716.16	5
256	0.00, 4.05, 30.43, 127.39, 369.98, 728.97, 1101.63, 1428.50, 1684.00	6
512	0.00, 4.55, 34.93, 152.42, 497.38, 1150.17, 1941.18, 2705.92, 3363.67, 3875.17	6
1024	0.00, 5.05, 39.43, 177.13, 619.31, 1667.37, 3214.68, 4876.81, 6426.01, 7745.51, 8769.01	7